

Monday 8-17-2026

Week 1

Day 1

Define the cyber security home lab purpose, concept, scope, and capabilities

Purpose

The purpose of this project is to design and build an automated cybersecurity home lab that provides a safe, isolated environment for learning, testing, and practicing cybersecurity skills. The lab will simulate real-world enterprise network environments, allowing the analyst to perform vulnerability assessments, security monitoring, incident response, and system hardening without affecting production systems.

Concept

Cybersecurity portfolio focused on hands-on network security, vulnerability assessment, packet analysis, reconnaissance, controlled exploitation, and security remediation through practical lab projects.

Project Scope

In Scope

The project includes:

- Laptop host computer
- Oracle VirtualBox
- Kali Linux virtual machine
- Vulnerable virtual machine (Metasploitable 2)
- Virtual networking configuration
- Vulnerability scanning
- Network traffic analysis
- Basic penetration testing
- Security documentation
- Automation using Bash and Python

- GitHub repository management
 - Monday.com project management
 - Weekly progress reporting
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Out of Scope

The following activities are **not** included in this project:

- Attacking public or production systems
- Cloud infrastructure (AWS, Azure, Google Cloud)
- Physical networking equipment
- Malware development
- Social engineering campaigns
- Wireless penetration testing
- Enterprise-scale Security Operations Center (SOC) deployment
- Commercial SIEM implementations

Cybersecurity Home Lab Capabilities

1. **Network Monitoring** — Monitor communication between virtual machines and observe network activity.
2. **Packet Capture & Analysis** — Use Wireshark to capture, filter, and analyze network packets.
3. **Network Traffic Analysis** — Examine protocols, connections, TCP/IP traffic, and suspicious network activity.
4. **Reconnaissance & Enumeration** — Discover hosts, open ports, services, and other information about the target system.
5. **Vulnerability Assessment** — Identify and evaluate vulnerabilities on the Metasploitable 2 target.
6. **Controlled Exploitation** — Safely practice authorized exploitation techniques within the isolated lab.
7. **Post-Exploitation Analysis** — Examine activities and network evidence following a controlled compromise.
8. **Security Analysis** — Analyze vulnerabilities, network behavior, and security findings.
9. **Remediation Testing** — Test security improvements and determine whether vulnerabilities or attack paths have been addressed.
10. **Security Reporting** — Document testing methods, findings, evidence, risks, and recommended improvements

